

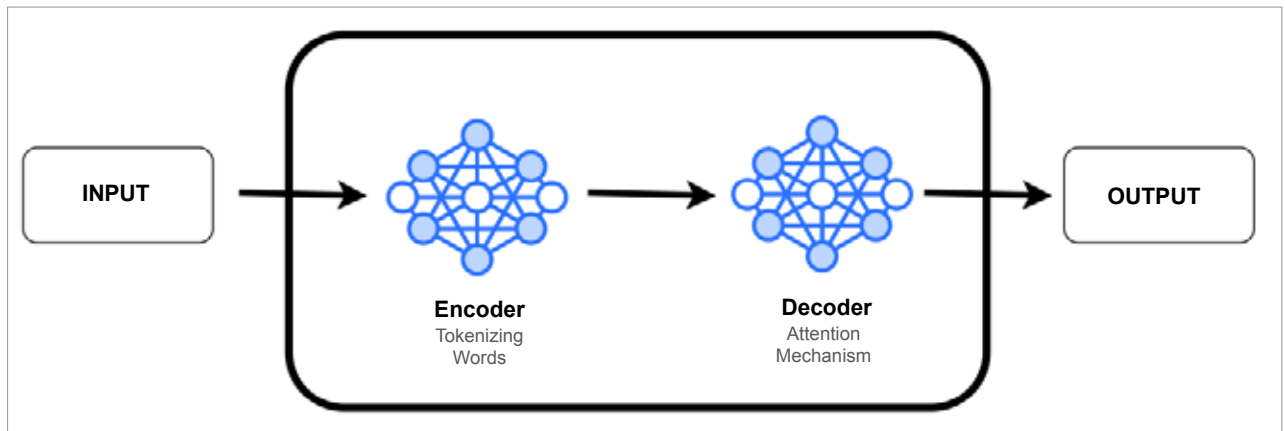


Figure 1: Key components of WormGPT

tailored explicitly for applications in natural language processing (NLP).

- Its training leverages the extensive Pile dataset, and its accessibility is facilitated through Mesh Transformer JAX.
- GPT-J showcases proficiency in generating English text in response to prompts and has demonstrated commendable performance across a spectrum of zero-shot downstream tasks.
- GPT-J is an open source counterpart to GPT-3, standing out as the most extensive publicly available GPT-3-style language model at its introduction.
- Its fundamental operation involves predicting the subsequent token in a given text string, rendering it versatile for diverse applications within NLP.

Working of WormGPT

- In contrast to ChatGPT, WormGPT functions as an unsupervised learning system, utilising deep learning algorithms and NLP to craft intricate conversational models that mirror human speech patterns.
- Due to the absence of safety measures and security protocols, WormGPT is vulnerable to exploitation by cybercriminals for generating sophisticated malicious content.
- The operational process of WormGPT includes word tokenisation, converting

words into numerical representations known as word embeddings, and utilising an attention mechanism for text generation.

- Its capabilities encompass generating text that resembles human speech, crafting phishing emails, and exploiting vulnerabilities to carry out malware attacks.
 - WormGPT is predominantly utilised for illicit purposes such as hacking and perpetrating payment scams.
 - Accessible through the dark web, WormGPT is specifically tailored for malicious activities, operating without ethical or legal constraints.
- The WormGPT model operates with two key components: the encoder and decoder blocks.

- **Encoder block:** This transforms input text into an internal representation within the model. It comprises multiple layers, each executing specific operations on words and phrases in the input text, including tokenisation and processing.
- **Decoder block:** Utilising representations from the encoder block, the decoder block employs an attention mechanism to generate output text. This mechanism, crucial for linking encoder and decoder blocks, enhances WormGPT's comprehension of word relationships in input and output texts. Leveraging internal

representations, the model generates output, allowing comparisons between new inputs and past outputs for predictions in a conversation.

The free version of WormGPT is reported to be semi-censored, and the uncensored version is available for purchase on platforms such as Telegram and Discord. The pricing plans for WormGPT range from 100 Euros for a monthly plan and 550 Euros for a yearly plan, to 5000 Euros for a private setup with enhanced security.

Usage of WormGPT as an ethical hacking tool

Using WormGPT for ethical hacking is a contentious issue, as it is primarily designed for malicious activities and lacks ethical boundaries. However, it's essential to consider that 'every coin has two sides'. While WormGPT is predominantly used for illegal activities, there might be potential ethical use cases. Here are some methods through which WormGPT could be used in an ethical hacking perspective.

- **Vulnerability assessment and penetration testing:** WormGPT could simulate and identify vulnerabilities in systems and applications. By generating attack scenarios and testing the resilience of systems, ethical hackers can proactively identify and address security weaknesses.